TRAFFIC & ACCIDENT INTELLIGENCE SYSTEM

Data Analytics Project Report

Project Area	Traffic, Accident & Emergency Event Analytics
Primary Data Source	911 emergency-event JSON dataset
Database	SQLite relational analytics model
Analytics	Python / SQL / dashboard-oriented analysis
Dashboard Focus	KPIs, trends, severity, hotspots, response and risk

Prepared as a structured academic project report based on the supplied Traffic & Accident Intelligence dashboard concept and the supplied 911 event data.

1. Abstract

The Traffic & Accident Intelligence System is a data analytics project designed to organize emergency-event data and convert it into meaningful traffic and accident insights. The supplied event data contains geographic coordinates, timestamps, township, ZIP code, address and incident title/description. Example records include traffic vehicle accidents, disabled vehicles, EMS incidents and fire incidents. The proposed system stores the event data in a structured SQLite model and produces analytical views such as incident trends, severity distribution, accident categories, geographic hotspots and response-oriented metrics.

The dashboard concept presents an operational overview with KPI cards, trend charts, traffic-volume analysis, hotspot visualization, critical incidents and an insights summary. The project therefore combines data preparation, relational database design, SQL querying, Python-based analytics and dashboard presentation into one end-to-end analytics workflow.

Important note: the supplied JSON is an emergency-event feed rather than a complete traffic sensor dataset. Therefore, measures such as traffic volume, response time, weather impact and risk scores should be treated as derived/enriched fields unless corresponding source data is added. The report keeps this distinction explicit.

2. Problem Statement

Emergency and traffic-related events can occur across many locations and time periods. When such records remain in a flat event format, it is difficult to efficiently compare locations, identify recurring incident patterns, analyze time-based trends and prioritize high-risk areas. A structured analytics system is required to transform raw event records into organized information that can support monitoring and decision-making.

3. Objectives

1. Store emergency and traffic-event records in a structured SQLite database.
2. Clean and organize geographic, temporal and categorical information.
3. Separate reusable dimensions such as location, date, time and category from incident facts.
4. Use SQL queries to calculate counts, trends, distributions and location-based summaries.
5. Use Python for data preparation, transformation and analytical calculations.
6. Present results through an interactive-style dashboard containing KPIs and visual analytics.
7. Identify incident hotspots and high-frequency locations using latitude and longitude.
8. Create a foundation for future risk scoring and real-time monitoring.

4. Data Source and Available Fields

The supplied 911 JSON data contains event-level records. The records shown in the supplied data include fields such as latitude, longitude, description, ZIP code, title, timestamp, township, address and an event flag. For example, records contain locations such as intersection addresses and titles such as 'Traffic: VEHICLE ACCIDENT -', 'Traffic: DISABLED VEHICLE -', EMS events and Fire events.

Field	Purpose in the project
lat	Geographic latitude for mapping and hotspot analysis
lng	Geographic longitude for mapping and hotspot analysis
desc	Detailed event description / source context
zip	ZIP/postal location grouping where available

Field	Purpose in the project
title	Primary event title and source category
timeStamp	Event date and time; used for trend and time analysis
twp	Township/local administrative area
addr	Event address/intersection
e	Source event flag

5. Proposed Data Model

A normalized/star-style model is recommended for SQLite. The central FACT_INCIDENT table stores individual events, while dimension tables store reusable location, category, date, time and station information. Derived risk analysis can then aggregate incidents by location and time. This design reduces duplication and makes SQL analytics easier.

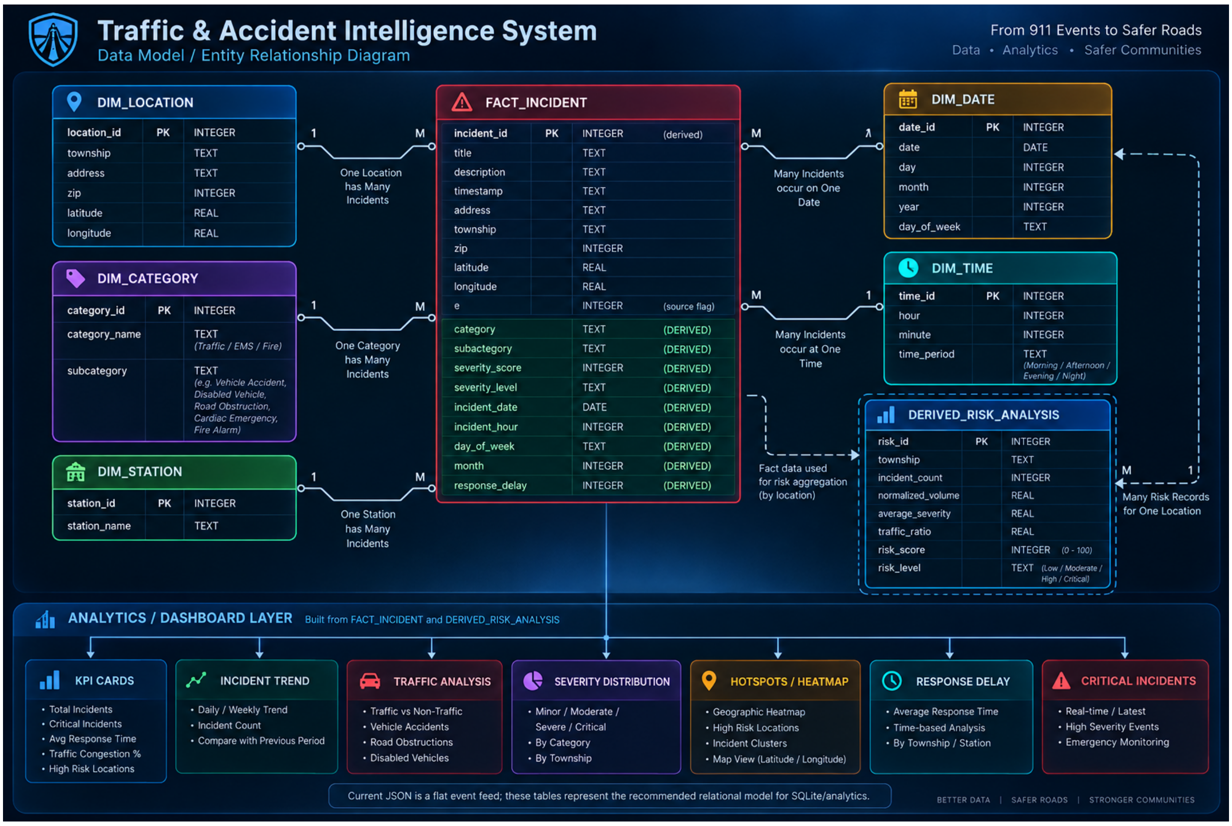

Figure 1. Recommended relational/data-model diagram for the Traffic & Accident Intelligence System.

5.1 Core Tables

Table	Important fields	Role
DIM_LOCATION	location_id, township, address, zip, latitude, longitude	Reusable geographic information
DIM_CATEGORY	category_id, category_name, subcategory	Incident classification
DIM_STATION	station_id, station_name	Emergency/service station reference
DIM_DATE	date_id, date, day, month, year, day_of_week	Date-based analysis
DIM_TIME	time_id, hour, minute, time_period	Hour/day-period analysis
FACT_INCIDENT	incident_id, title, description, timestamp, location/category/date/time keys/derived fields	Central event/fact table
DERIVED_RISK_ANALYSIS	risk_id, location/township, incident_count, average_severity, traffic_ratio, risk_score, risk_level	Aggregated risk analysis

6. System Workflow

Step	Stage	Description
1	Raw JSON	Collect 911 emergency-event records
2	Data Cleaning	Handle missing values, standardize timestamps and text

Step	Stage	Description
3	Classification	Derive category/subcategory from event titles
4	SQLite Storage	Load normalized records into relational tables
5	SQL Analytics	Calculate counts, trends, location summaries and distributions
6	Python Analytics	Transform data and generate analytical metrics/visualizations
7	Risk Layer	Aggregate indicators by location/time for risk scoring
8	Dashboard	Display KPIs, charts, hotspots and critical incidents

7. Analytics and Dashboard Design

Dashboard component	Analytical purpose
Total Accidents	Overall count of classified accident records
Critical Accidents	Count of events classified into the highest severity group
Traffic Incidents	Count of traffic-related incidents
High Risk Locations	Locations/townships meeting the selected risk criteria
Average Response Time	Response metric when response-time data is available
Traffic Congestion	Congestion indicator when traffic-volume/sensor data is available
Accident Trend Over Time	Daily/weekly incident count trend
Traffic Volume Analysis	Time-based vehicle volume; requires traffic-volume data
Accident Hotspots	Geographic clustering using latitude/longitude
Severity Distribution	Share of incidents by severity level
Accident Type Distribution	Share by incident category/subcategory
Road Condition Impact	Comparison by road-condition data if available
Critical Incidents	Latest/high-priority events for operational monitoring

7.1 Example SQL Analysis Questions

• How many traffic-related incidents occurred per township?

• Which locations have the highest number of vehicle accidents?

• What are the busiest hours for traffic incidents?

• How does the incident count change by day of week?

• What are the most common incident titles/subcategories?

• Which geographic points contain repeated incidents?

• How many critical incidents occur within each selected date range?

8. Risk Analysis Concept

The risk layer is a derived analytics component rather than a direct field in the supplied JSON records. A location-level risk score can be calculated from measurable indicators such as incident frequency, severity and recency. If traffic-volume data is later integrated, a traffic-normalized accident ratio can also be included.

Indicator	Example contribution
Incident frequency	More incidents at a location increase risk
Severity	Severe/critical incidents receive greater weight
Recency	Recent incidents can receive higher weight
Traffic exposure	Accidents normalized by vehicle volume when available
Location clustering	Repeated nearby incidents increase hotspot strength

9. Technology Stack

Layer	Suggested technology
Data source	JSON / CSV event data
Database	SQLite
Database access	Python sqlite3 module
Data processing	Python
Analytics	SQL + Python
Visualization	Python visualization libraries / dashboard tool
Dashboard	Web dashboard or BI-style dashboard
Geospatial view	Latitude/longitude-based map or heatmap

10. Limitations and Data Requirements

• The supplied event data is primarily an emergency-event feed and does not by itself provide complete traffic-sensor measurements.

• Response time is not directly present in the supplied records; it must come from another operational source or be explicitly marked as unavailable.

• Weather and road-condition impact require corresponding weather/road-condition datasets.

• Traffic congestion and traffic volume require sensor, GPS, road-counter or equivalent data.

• Severity levels and risk scores are derived classifications unless a verified severity/risk field is supplied.

• Missing ZIP values occur in the source data, so geographic analysis should handle missing values safely.

11. Future Scope

• Integrate live traffic sensor and congestion data.

• Add weather API data and road-condition observations.

• Add emergency-service dispatch and response-time data.

• Implement automated hotspot detection.

• Build a live web dashboard with filters and drill-downs.

• Add scheduled data ingestion and automatic database refresh.

• Introduce predictive modeling only after sufficient labeled historical data is available.

• Deploy role-based operational views for administrators and emergency operators.

12. Conclusion

The Traffic & Accident Intelligence System provides an end-to-end framework for converting 911 event records into structured, queryable and visual analytics. The recommended SQLite data model separates event facts from reusable dimensions such as location, date, time and category. SQL and Python can then be used to calculate incident trends, distributions and hotspots, while a dashboard presents the results in an operational form. The supplied data already supports meaningful geographic, temporal and categorical analysis; additional traffic, weather, road-condition and response datasets can extend the system into a richer real-time intelligence platform.

Report status: Academic project design and analytics blueprint. Derived metrics should be validated against their underlying source datasets before being presented as real operational measurements.